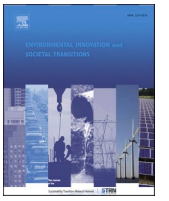
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Play, games, and gamification to support sustainability transitions: a scoping review and research agenda

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ABSTRACT

Transitioning towards more sustainable systems is crucial in an era where carbon emissions continue to rise and planetary boundaries have been exceeded. However, traditional methods for public engagement with institutionally, technically, and ecologically complex issues have shown limited results in motivating change. In response, play, games, and gamification promise to support sustainability transitions by engaging diverse stakeholders and promoting psychological and behavioral changes. Despite these promises, we lack systematic knowledge of whether such techniques, which typically leverage individual motivation and agency, can bring together stakeholders in diverse areas. Thus, we conducted a scoping review of 86 empirical outputs, which revealed untapped potential to transform practices and support large-scale transitions. Based on our findings, we propose three future agendas. First, our contextual agenda encourages clearly anchored and critical sustainability conceptualizations, engaging practitioners for resilient transitions, considering emotions, supporting technical and ecological transitions, and expanding beyond the West and the local scale. Second, for design, we suggest user-centered approaches, customizable interventions, a broader conceptualization of play, combining game and goal types, including more immersive game elements and technologies, and leveraging tools and methods beyond gamification. Third, our empirical agenda calls for larger sample sizes and longer studies, contrasting conditions, employing mixed methods, exploring double and triple loop learning, displaying critical reflexivity, and reporting clearer causal links and experiences. Future research may utilize these points to explore and leverage the potential of playful and gameful interventions in changing transition dynamics, environments, and actors for practice and regime-level system transformations.

1. Introduction

As carbon emissions continue to rise globally and localized reductions are slow (Friedlingstein et al., 2024) and most planetary boundaries have been crossed (Richardson et al., 2023), it is crucial to abandon incumbent systems in favor of more sustainable ones (Sengers et al., 2019). However, attaining sustainability is a challenging endeavor—meeting "the needs of all within the means of the planet" (Raworth, 2017, p. 9) involves and connects areas "such as technology, the economy, institutions, behaviour, culture, ecology and belief systems" (Rotmans et al., 2001, p. 2).

Accordingly, the research field of sustainability transitions (STs) has seen over two decades of evolution. STs affect socio-technical,

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socio-institutional, and socio-ecological systems in complex ways—transitions are long-term processes with uncertain outcomes that involve multiple domains, actors, and innovations assailing path dependency and otherwise stable systems (Loorbach et al., 2017). STs concern diverse values and preferences, even among those who agree that transitions are desirable (Köhler et al., 2019), and impact many stakeholders in both problematic and beneficial ways (Suboticki et al., 2023).

Given the above, public *engagement*, understood as direct participation in decision-making and policy processes (Rowe and Frewer, 2005), is relevant for equitable democratic planning, regulation, and implementation (Suboticki et al., 2023). Thus, many ST engagement initiatives aim to promote stakeholder *participation*. More broadly, the term "engagement" is also used in the sustainability context to describe a personal state of connection with an issue "concurrently comprising cognitive, affective and behavioural aspects' (Lorenzoni et al., 2007, p. 446), so public ST engagement interventions also promote, e.g., learning and attitudinal change.

For actors to better engage with STs, acquiring and making sense of information is crucial (Stam et al., 2023). In STs, engagement through *learning* occurs in several ways. One essential approach to ST learning, termed "learning by doing" (Van Poeck et al., 2020), includes methods such as discussions to reach a common understanding and iterative processes of action and reflection (Van Mierlo and Beers, 2020). Other approaches include social learning, which situates understanding within social units or communities of practice (Reed et al., 2010), and transformative learning, which criticizes and shifts one's frames of reference (Quang and De Wit, 2020). In addition to different learning approaches in STs, scholars have also emphasized the criticality of different learning levels. Barth et al. (2023) describe three such levels: Single loop, where new information leads to doing things differently; double loop, where rules are reevaluated leading to doing different things; and triple loop, where tools are developed to support double loop transformations.

ST-relevant *behavioral engagement* is often understood through the lens of theories that observe everyday practices such as mobility or eating. These perspectives aim to transcend the limitations of centering individual agency (Shove, 2010) and consider both actors and their environments instead (Welch and Yates, 2018). Practice-anchored ST research asks "how variously sustainable practices come into existence, how they disappear and how interventions of various [practice] forms may be implicated in these dynamics" (Shove and Walker, 2010, p. 471). From this perspective, "interventions go on within, not outside, the processes they seek to shape" (Shove, 2010, p. 1278).

Engagement is further complicated by the role of *emotions* in STs, which affect innovation adoption (Valor et al., 2022). Recognizing how emotions affect learning processes is crucial for understanding the role that motivation, cognitive skills, and socio-emotional developments play in shaping sustainable practices (Thelin, 2024). For example, a positive emotional commitment is said to be required for willful change (Martiskainen and Sovacool, 2021). Environments that elicit positive emotional experiences and enhance individual motivation can reinforce personal goals, agency beliefs, and the acquisition of skills (Michaelides, 2008; Naude et al., 2014).

As seen above, stakeholder participation, learning, affect, and behavior are all relevant to ST engagement. Against this backdrop, the present study explores how gameful and playful interventions motivate and support these aspects. As contemporary culture becomes gradually more gamified, i.e., as play and games become intertwined with a host of different systems and activities (Hamari, 2019), an increasing number of systems intentionally use game design and playful experiences to bring psychological and behavior change in contexts ranging from education to healthcare (Hamari et al., 2014; Hamari, 2019). Although serious games and gamification have had slightly different points of departure traditionally (Landers, 2014), they are currently considered as belonging under a shared conceptual umbrella, since both seek to cause gameful engagement (Douglas and Brauer, 2021; Koivisto and Hamari, 2019; Krath et al., 2021; Rajanen and Rajanen, 2019). Thus, "gamification" has been proposed as the umbrella term to describe the result of transforming activities or systems to afford game-like experiences and support psychological or behavior change, by digital or analog means (Hamari, 2019). Beyond these approaches, we also posit that play activities, or those "designed to generate engagement more than a relevant consequence" (Masek and Stenros, 2021, p. 23), can also be understood as a form of gamification, which is why we will primarily use this word.

The relationship between games and sustainability is complex. On the one hand, the unbridled growth of the gaming sector is not aligned with an environmentally sustainable future and the gaming industry itself is in dire need of a transition (Abraham, 2022). On the other hand, games and gamification have been utilized to promote both engagement with climate change (Fernández Galeote et al., 2021) and environmental education and behavior (Boncu et al., 2022; Miao et al., 2022). Gamification has been applied to ST-relevant domains such as sustainable consumption (Guillén et al., 2021a), energy (Beck et al., 2019; Johnson et al., 2017), mobility (Avril et al., 2024), agriculture (Dernat et al., 2025), and, more broadly, sustainable social innovation processes (Guillén et al., 2021b). As the number of serious games for ST education has increased sharply (Stanitsas et al., 2019), research has found that they can represent ecological, economic, and political structures (Foppe and von Wehrden, 2024), but they do not typically offer a balanced view of STs that considers environmental, economic, and social aspects at once (Stanitsas et al., 2019). Furthermore, the imaginaries contained in digital games about energy tend to be simplistic, rooted in hegemonic views of energy production, and favor gradual over radical change (Wagner and Gałuszka, 2020).

More deeply, the ontology of STs is generally seen as a set of radical and complex changes (Köhler et al., 2019). This ontology intersects with gamification's own ontology, as a tool for psychological and behavior change (Hamari, 2019). Gamification often aims to educate stakeholders about complex sustainability issues (Fernández Galeote et al., 2021), but a degree of simplification seems necessary to model reality in a game or promote change. Whereas a long-term view and uncertainty are tenets of STs, gamification is constrained by resources and attention spans, and only seems to favor uncertainty as long as it supports the play experience and the desired (usually personal) change. This begs the question of whether gamification should be seen as just another "green" solution for short-term, small-scale deployment supported by landscape-level trends, or as a force capable of permeating other innovations and underpinning regime shifts. Further, from the practice theory perspective, we may question *whether and how gamification is used to*

transform practices, e.g., by targeting the user's motivation, resignifying action, and altering environmental cues, incentives and affordances; or whether its focus on ST-related change is mostly disconnected from the process it aims to shape.

The ontological complexity of STs is mirrored in current epistemological and methodological debates within the field (Geels, 2022). As STs are heterogeneous, multi-dimensional, large-scale, long-term, and their relevant components fluctuate constantly, critical realism has been proposed as an adequate philosophy of science for them, as its explanations "provide an account of how outcomes are generated by actors and causal mechanisms in structural contexts" (Geels, 2022, p. 4). In particular, Geels (2022) proposes an agentic-processual approach to critical realism, which explicitly considers actors and social interactions as causal agents conditioned by structures. Considering the special epistemological and methodological needs of ST research and the fact that sustainability gamification is typically applied as a motivational, educational, and behavior change tool at the small scale, an appropriate question is *whether and how those agentic changes (*e.g.*, in awareness, motivation, or behavior) feed into larger transition dynamics.*

An inquiry in such broad directions requires a map of the territory. Multiple literature reviews have focused on research areas adjacent to or part of gamified STs, such as serious games for sustainability education (Ahmadov et al., 2024) and gamification for sustainable behavior change (Guillén et al., 2021a; Miao et al., 2022), including in specific areas such as energy consumption (Beck et al., 2019; Johnson et al., 2017) or transport (Avril et al., 2024). Other reviews have surveyed the state-of-the-art of gamification for climate change engagement (Douglas and Brauer, 2021; Fernández Galeote et al., 2021) and sustainable energy, water, food, mobility, and education more broadly (Boncu et al., 2022). Therefore, different areas of education and action relevant to STs have been covered extensively. However, analytical lenses aligned with STs were not used in these reviews.

We aim to fill this critical gap by exploring the extent to which gamification has been used as part of ST processes and to offer unique insights by analyzing interventions explicitly as part of STs, rather than as tools to promote sustainability in specific contexts (e.g., formal education, household consumption) or areas (e.g., energy, food, mobility). We approach STs as ontologically distinct processes with their own epistemological and methodological features. In this way, we direct our research scope to empirical studies with explicit contributions to STs or those using closely related systemic terms. While a few reviews have explicitly targeted serious games and STs (Dernat et al., 2025; Foppe and von Wehrden, 2024; Stanitsas et al., 2019; Wagner and Gałuszka, 2020), they tend to focus on serious games and their features. Conversely, this review includes multiple types of activities rooted in play and reports on the research interventions' contexts, methods, and outcomes in addition to game system features.

Based on the above, we conducted a scoping review in the area of play, games, and gamification in STs. This method is suitable when broad questions are posed (Tricco et al., 2018) in a complex research area that has not been comprehensively reviewed before (Mays et al., 2001). In addition to mapping out the research and summarizing its findings, we aimed to identify gaps and opportunities that we articulated through a research agenda (Arksey and O'Malley, 2005).

Four research questions guide the review:

- RQ1. In what contexts and populations have gamified interventions for STs been applied?
- RQ2. What are the characteristics of the gamified interventions used in STs?
- RQ3. What scientific methods have been used to evaluate the impacts of gamification in engaging people with STs?
- RQ4. What have been the results of applying gamification to engage people with STs?

This review's contribution is double. On the one hand, its holistic view of engagement rooted in playfulness will support educators, designers, and researchers in the growing field of gamified STs in planning and conducting more effective and novel interventions. On the other hand, its use of ST-specific lenses sheds light on the current state of gamification not just as a tool for sustainability learning, persuasion, and behavior but more broadly as a vehicle to advance transitions themselves, where much work is still needed to holistically transform practices and boost large-scale dynamics. Beyond the agenda, which will broadly inform gamified interventions for sustainability, the data extraction matrix included in the supplemental file provides a detailed analysis of all the research outputs.

2. Methods

This review follows the PRISMA-ScR guidelines (Tricco et al., 2018), a rigorous and transparent method for comprehensively searching, selecting, extracting, and analyzing research outputs relevant to our four research questions. To ensure systematicity, we drafted a review protocol accessible to all researchers tasked with extracting and analyzing the data. The protocol presents the research questions and defines how relevant studies should be identified and selected, and how the data is to be charted (Arksey and O'Malley, 2005).

2.1. Study identification and selection

The Scopus database was chosen since it indexes influential journals in the field of sustainability transitions, such as Energy Policy, Research Policy, Technological Forecasting and Social Change, Technology Analysis and Strategic Management, and Environmental Innovation and Societal Transitions (Zolfagharian et al., 2019). It has been the database of choice in ST reviews because it is the largest in the world (Martiskainen and Sovacool, 2021) and the most comprehensive one in the field (Stam et al., 2023). Scopus also covers the main sources in the area of gamification and serious games (e.g., ACM Digital Library, IEEE Xplorer, ScienceDirect, and Springer Link).

The search string used covers multiple terms relevant to STs including "transformation," which can be used together with "transition" to refer to the outcome of a process of change (Hölscher et al. 2018). Keywords were defined through previous field knowledge and refined through multiple database searches to ensure a balance between result comprehensibility and relevance. The final string is

as follows:

TITLE-ABS-KEY (("sustainab trans*" OR "socio-technical change*" OR "sociotechnical change*" OR "sociotechnical trans*" OR "socio-technical trans*" OR "socioecological trans*" OR "socio-ecological trans*" OR "ecological trans*" OR "energy trans*" OR "food trans*" OR "mobility trans*" OR "sustainab* practice*" OR "experiment* in living" OR "voluntary simplicity" OR "simple living" OR "low impact living" OR "transition town*" OR "grassroots innovation*" OR "inclusive innovation*" OR "frugal innovation*" OR "niche innovation*" OR "niche experiment*" OR "socio-technical experiment*" OR "grassroots experiment*" OR "transition experiment*" OR "sustainability experiment*" OR "living lab*" OR degrowth OR post-growth OR post-capitalis* OR postcapitalis* OR prefigurati* OR communalis* OR "interstitial revolution*" OR ecosocial* OR "buen vivir" OR solarpunk OR hopepunk OR salvagepunk OR afrofuturis* OR "seed bombing" OR "guerrilla gardening" OR "dumpster diving") AND (humor* OR humour* OR artist* OR playful* OR gameful* OR gamif* OR "game-based" OR "board game*" OR "card game*" OR "video game*" OR videogame* OR "digital game*" OR "mobile game*" OR "online game*" OR "computer game*" OR "serious game*" OR "educational game*" OR "role-playing game*" OR "green game*" OR "environmental game*") AND NOT ("we hope" OR "our hope")) AND (LIMIT-TO (LANGUAGE, "English") OR LIMIT-TO (LANGUAGE, "Spanish") OR LIMIT-TO (LANGUAGE, "French") OR LIMIT-TO (LANGUAGE, "Italian") OR LIMIT-TO (LANGUAGE, "Portuguese") OR LIMIT-TO (LANGUAGE, "German")) AND (LIMIT-TO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE, "cp") OR LIMIT-TO (DOCTYPE, "ch"))*

The initial search was conducted in March 2024, followed by a second search in February 2025 to include all outputs published in 2024. The string was applied to the outputs' title, abstract, and keywords. We limited our search to journal articles, book chapters, and conference articles. The languages included were limited to those that the research team in charge of the study screening process could read: English, Spanish, French, Italian, Portuguese, and German. The final selection consisted of 80 outputs in English, four in Spanish, two in French, and one in Italian. There was no limitation related to the publication date.

In terms of content, articles were included if they complied with three inclusion criteria. First, involving playful methods. We conceptualize play as an action or activity rooted in a playful mindset (Stenros and Montola, 2024). A playful mindset is one in which people "prioritize engagement over external consequence, realness, or convention" and thus play activities are those "designed to generate engagement more than a relevant consequence" (Masek and Stenros, 2021). While relevant consequences remain important in gamified contexts, we understand that playful methods are chosen because the engaging process they offer is expected to support its outcomes. Thus, we included play, games, gamification, and interactive and participatory forms of art engaging non-artists, from gamified apps to "practices … informed by principles such as self-imposed constraints, bricolage, or collective creativity" (Op de Beke et al., 2024). Accordingly, the search string includes a series of terms that exist at the intersection between sustainability transitions and playfulness. These span playful "green" practices such as guerrilla gardening, seed bombing, and dumpster diving (Op de Beke et al., 2024). It also encompasses artistic genres such as solarpunk, hopepunk, and salvagepunk, which explore sustainable futuring and speculative possibilities through art and creativity (Mancuso, 2021; Wilson, 2023). Similarly, Afrofuturism offers artistic visions of an often sustainable, high-tech future beyond colonial exploitation, which can serve as inspiration for real-life transitions (Osman, 2021). Other understandings of words such as "game" not based in a playful mindset, e.g., as it is used in economic game theory (e.g., Stackelberg games) were excluded.

Second, engaging with sustainability transitions. This means that the study contributed to "a set of connected changes, which reinforce each other but take place in several different areas, such as technology, the economy, institutions, behaviour, culture, ecology

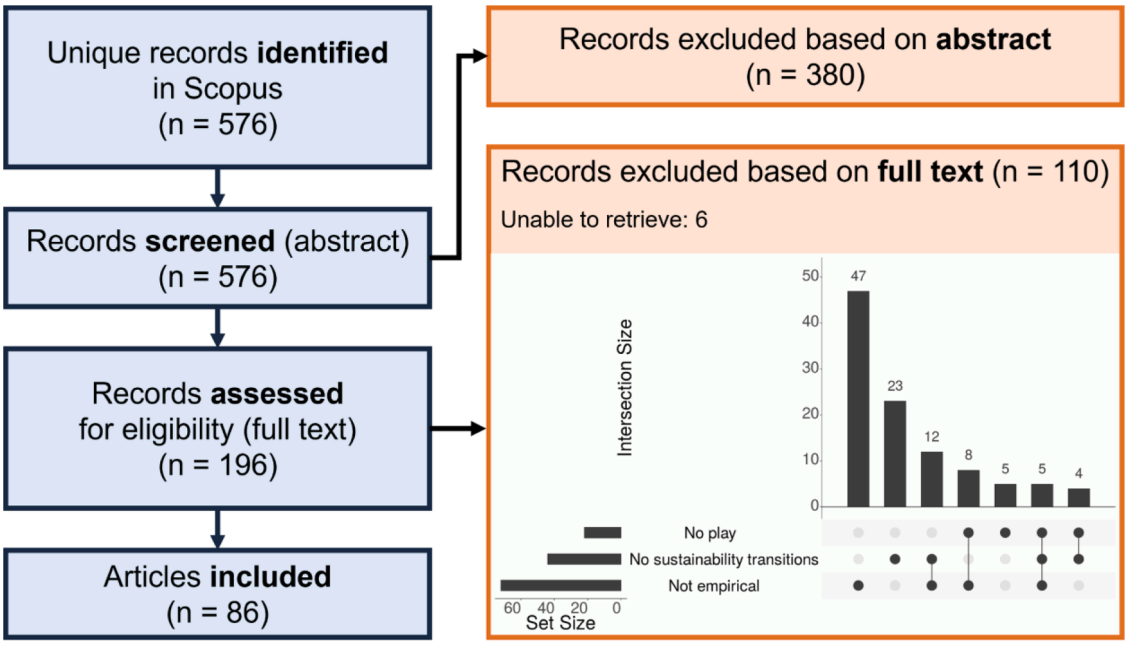


Fig. 1. PRISMA flowchart. The UpSet plots in this review were created using UpSetR (Lex et al., 2014).

and belief systems" (Rotmans et al., 2001, p. 2) with the goal of abandoning an incumbent system in favor of one deemed more sustainable (Sengers et al., 2019), i.e., one that can "persist indefinitely without a decline in the resource base or in the social welfare it delivers" (Walker and Salt, 2012, p. 216). We include socio-technical, socio-institutional, and socio-ecological systems (Loorbach et al., 2017). In the context of this review, the use of play, games, and gamification could affect large systems or be part of smaller interventions relevant to transitions such as sustainability experiments (Sengers et al., 2019). Here, we have included transition movements and strategies that have notable artistic or engagement-prioritizing aspects (e.g., joy, fun, happiness, or motivation-oriented), and therefore are particularly suited for a study focused on playful and gameful sustainability transitions. For instance, postcapitalist and post-growth perspectives explore socioeconomic models that are both environmentally sustainable and support human motivations beyond seeking economic profit and material consumption and accumulation, e.g., through the use of measures such as the Gross National Happiness (GNH) index (Kallis, 2015). These are also related to degrowth, whose scholars have called for the search of "more meaningful and less destructive forms of happiness" (Schmelzer et al., 2022), and "buen vivir," which proposes alternatives to hegemonic forms of development built upon a respectful relationship with nature and considers concepts such as happiness in its transitions (Gudynas, 2011).

Third, the study had to report empirical results on the role of playful and gameful interventions in engaging participants with sustainability transitions.

The process of output screening and selection was carried out by Authors 1 and 2, with the involvement of Author 3 when a third point of view was required. The process is illustrated in Fig. 1. The final list of 86 outputs is available in the supplemental file.

2.2. Data charting

The data extraction variables were defined in the protocol and put into a matrix for data extraction. The variables, together with their descriptions and references, can be found in the supplemental file. Data extraction was performed independently for all outputs, involving Author 1 and either Author 2 or 3, depending on the output's language. All three authors extracted social interaction and learning forms variables due to their level of abstraction and a higher probability of diverging personal interpretations. Wherever possible, verbatim explanations were extracted from the outputs during this process to ensure fidelity.

The data were extracted in batches. Authors 1 and 2 edited the protocol until they both found it clear. Then, they extracted the data from three articles as a trial. After this, they met and adjusted the variables and descriptions based on perceived issues. Data extraction continued in batches of five to 15 outputs as the analysts became more familiar with the process. At the end of each batch, the analysts met and agreed on a final value for each cell. Thus, both the output screening and the data collection followed a consensus-based approach that prioritized exploring and resolving disagreements (Boutron et al., 2019).

After reaching consensus, Author 1 created further inductive categories for those variables in which the data were too complex to summarize for readers (sustainability framing, problem framing, scale, system framing, duration, game goal, sample size, behavioral outcomes, and mediating experience; see supplemental file). Next, Author 1 wrote the article's first draft. The final classifications were shared with all co-authors, who contributed to editing the final manuscript. The results are reported in the following section.

3. Results

3.1. Bibliographic elements

In addition to the data answering our research questions, we collected bibliographic information to identify general trends in this research area. Fig. 2 shows the data starting in 2015, the first output's publication year. The year with the most outputs was 2024, which almost doubles 2023 and indicates a possible rising trend.

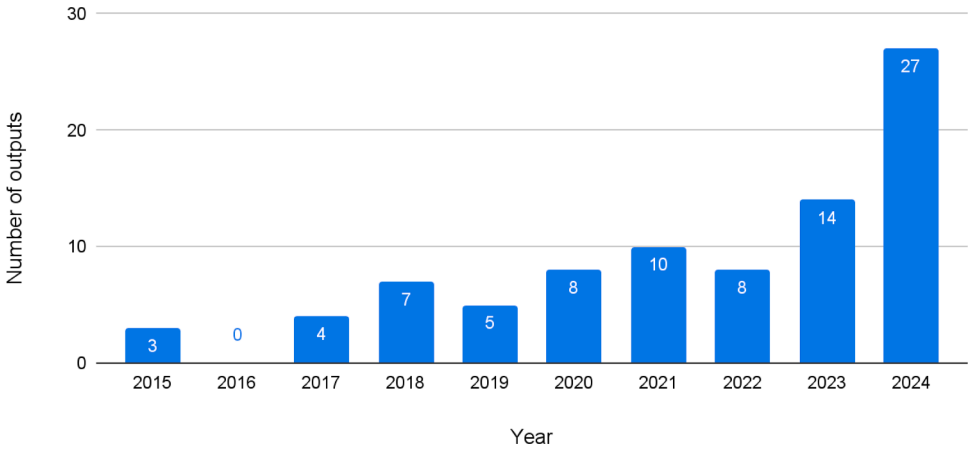


Fig. 2. Number of outputs by year.

The final sample comprises 57 journal articles, 21 conference papers, and 8 book chapters. The outputs were published in 67 venues. The venues with more than one output include the journals Sustainability (6), Sustainability Science (4), and Energies (3), the IOP Conference Series Earth and Environmental Science (3), the Journal of Cleaner Production (3), Agricultural Systems (2), the Annals of the Association of American Geographers (2), the Electronic Journal of e-Learning (2), the Proceedings of the European Conference on e-Learning (2), and the ACM International Conference Proceeding Series (2). Thus, most popular venues focus on sustainability and natural science rather than games or technology. An analysis of subject areas, as defined by Scopus, reveals that 42 outputs were published in venues within the Social Sciences category, followed by Environmental Science (26), Computer Science (20), Engineering (16), and Energy (16). Other categories had fewer than 10 outputs each. 17 outputs were in venues with no subject categories assigned. The most repeated categories suggest that this research area combines the social sciences, the environment, energy and other technological systems, and digital solutions.

Based on the author teams' institutions, 38 countries are represented in the sample, of which 18 register more than one output: The Netherlands (16), Italy (14), Germany (11), Spain (11), the US (10), Sweden (7), Switzerland (7), the UK (6), France (5), Greece (3), Ireland (3), Norway (3), Austria (2), Canada (2), China (2), India (2), Malaysia (2), and Romania (2), indicating a predominance of European countries.

3.2. Context and population

RQ1 refers to the context where the playful and gamified interventions were applied and to the participating populations. We begin this section by examining how the studies define sustainability and the problems they address with their interventions. We then proceed to the area of application and type of transition, followed by the intervention's location and scale, and the actors involved.

3.2.1. Sustainability framing

Our first contextual variable of interest is how the outputs framed sustainability, i.e., whether they were specific about the word's meaning, and where they anchored the concept. An inductive classification of the prevalent framing in each output led to three categories. First, 55 % of the outputs prioritized institutional views. These include definitions, frameworks, reports, and targets established by local and national governments and by supranational institutions such as the European Union, the United Nations, the World Health Organization, the Intergovernmental Panel on Climate Change, and the Paris Agreement. A clear example of this cluster is anchoring on sustainable development and the UN's Sustainable Development Goals.

A second category includes outputs that lacked an explicit definition of sustainability (23 %). These mentioned sustainable development without further referencing or contextualization, or indicated the need to develop renewable energies, reduce waste, engage in nature conservation, and address climate change through decarbonization and environmental footprint reduction in a general way.

Finally, the third category includes outputs that engage with alternative or critical framings (22 %), including opposition to neoliberal systems via affective and playful labor (A41), criticisms to UN concepts such as "self-made cities" (A35), emergent practice frameworks such as action-oriented research (A12), orientation towards the commons and socio-ecological transformation (A10), ecosophy (A54), eco-pacifism and decolonization (A64), symbiosis (A06), and more-than-human perspectives and other ontological and epistemological departures from mainstream ways of seeing the relationship between humans and the environment.

3.2.2. Problem framing

Outputs typically framed the need for playful or gamified techniques in their introduction sections. Typically, more than one reason was indicated. We collected evidence of these framings and inductively constructed the classification presented in Table 1. As expected in gamification, the main stated problems were related to behavior change and learning. However, many outputs listed the need to design, create, and imagine and thus favored the inclusion of more open-ended, participatory, and playful processes than education or persuasion. Despite the importance of attitudes, emotions, and constructs such as self-efficacy in behavior change theories and frameworks, these issues were less commonly referred to than behavior change and decision-making themselves.

3.2.3. Application area, transition type, and systemic connections

Areas of application were deductively grouped into energy (35 % of all outputs); agriculture, forestry, and other land use (AFOLU) (33 %); transport (22 %), industry (13 %), and buildings (IPCC, 2023). We found no outputs that addressed building emissions separately from energy consumption (electricity and heating). Still, of the 30 outputs that included energy, 24 referred to households

Table 1
Number and percentage of outputs by problem framing.

Problem to address	Output number and percentage
Behavior and decision-making	52 (60 %)
Understanding, awareness	50 (58 %)
Design, creation, visioning	24 (28 %)
Attitude, affect (e.g., anxiety), self-efficacy	14 (16 %)
Data elicitation	3 (3 %)
Ethical questions and imperatives	2 (2 %)
Gamification improvement	1 (1 %)

or local community action, and some interventions targeted future architects (A69) and civil engineers (A31). Apart from the outputs represented in Fig. 3, 11 did not clearly refer to a mitigation sector but to general design, creation, and visioning processes, while two included adaptation concerns.

Fig. 4 shows the outputs' foci and connections to other systems. A socio-institutional focus was found in most of them, while none were exclusively socio-ecological, i.e., focused only on environmental matters and resilience without tackling issues of infrastructure and social structures, cultures, and agency. In terms of connections, less than half stated the intervention's political, economic, and ecological implications together, while six outputs did not clearly report any.

3.2.4. Location and scale

The interventions described took place in 36 countries, plus two cases which involved global online participation. Of the countries represented, 16 were found in more than one output, most of them in Europe: The Netherlands (14), the US (10), Germany (9), Italy (10), Spain (9), Sweden (7), Switzerland (6), Greece, Ireland, Norway (3 each), and France, India, Malaysia, Poland, Portugal, and the UK (2 each).

In addition to the intervention's location, we recorded the geographic scale of the targeted systems, that is, the largest scale of change that the participants experienced. 27 % of the outputs referred to transitions at the building (e.g., household), institution (e.g., school), or single business (e.g., farm) level; 53 % referred to local practices and infrastructures; 16 % included transformations at the regional or national level; and 3 outputs referred to practices conducted at the international or global level. In temporal terms, all studies communicated the goal of affecting participants beyond the intervention's end. This is because they either (a) imparted skills or aimed to ingrain habits, for example when targeting students; (b) involved long-term technological (e.g., solar energy provision) or environmental change (e.g., soil regeneration); (c) contributed to achieving future environmental targets (e.g., 2030, 2040, 2050); (d) involved visioning and shaping the future; and/or (e) built gameful systems expected to be used in the future as well.

3.2.5. Actors involved

To better understand who took part in the interventions, we extracted data about player populations and classified them using the quadruple helix model: academia, governments, industry and businesses, and the public, including cultural and media actors as participants (Carayannis and Campbell, 2009). Fig. 5 presents data from player populations and non-players who had a prominent role in the interventions. In all cases except five (A10, A64, A71, A73, A86), academics were involved either as participants or as meaningful assisting actors.

3.3. Intervention

RQ2 focuses on intervention characteristics, including (a) the systems' general features—framing, medium, number of players, duration, relationships to real locations, involvement of larger structures and technologies; (b) game design aspects such as the game's

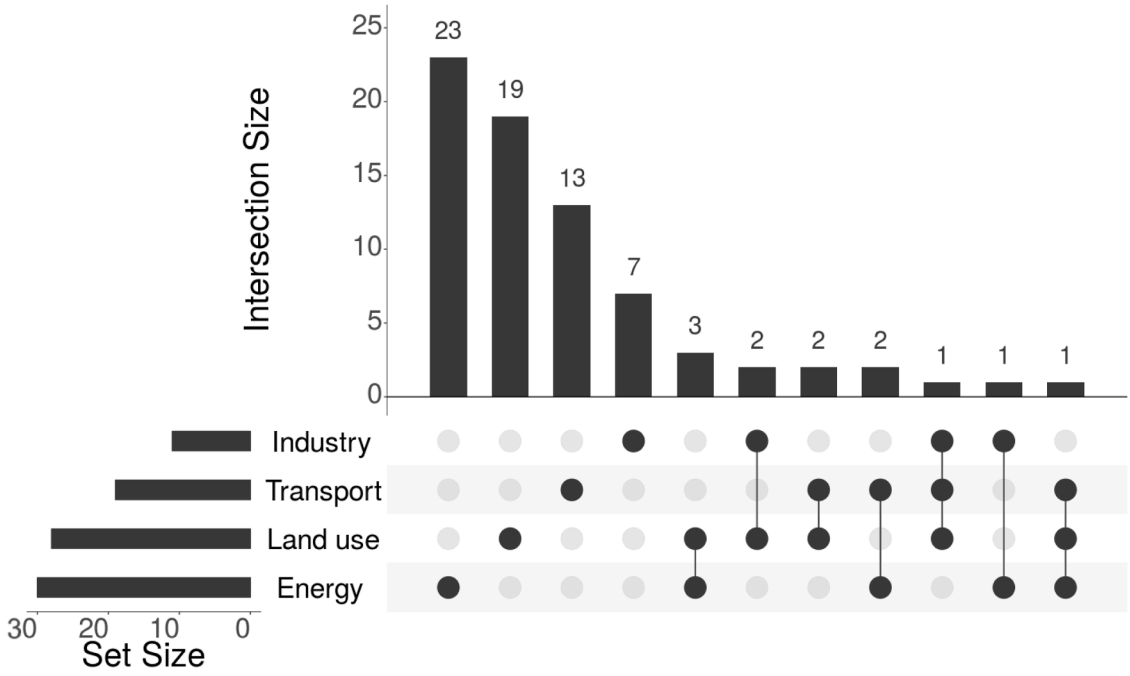


Fig. 3. Number of outputs by application area. Apart from these 74 unique outputs, 11 did not refer to a particular sector, and one referred to adaptation alone.

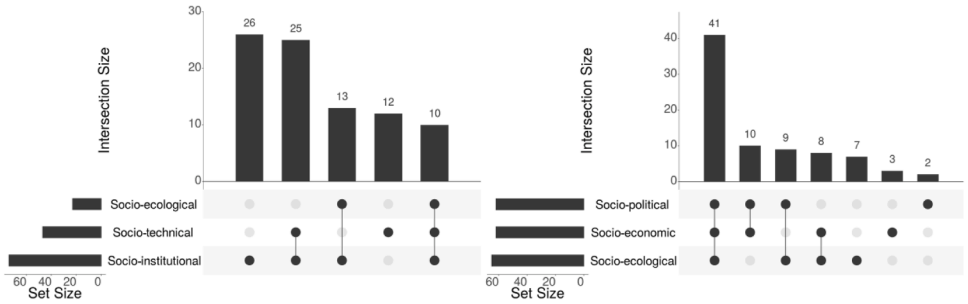


Fig. 4. Number of outputs by focus (left) and connections to other systems (right).

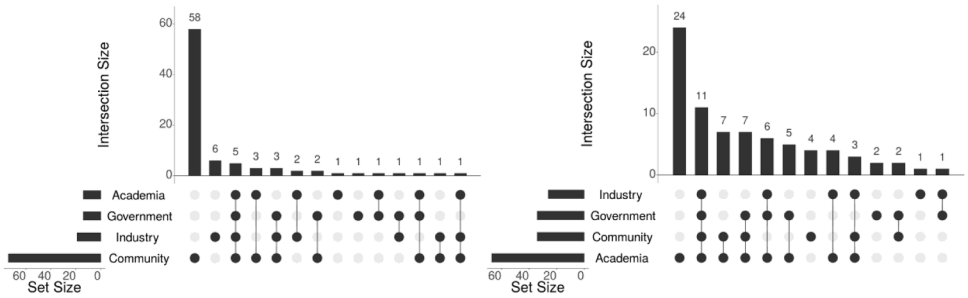


Fig. 5. Player populations (left) and assisting actor populations (right).

goal and gamification elements; and (c) other variables relevant to the transition at hand—the gamified system's innovation reach, targeted change type, and consideration of practices.

3.3.1. *General system characteristics*

Studies referred to the systems employed using multiple terms. On the one hand, 70 % referred to games, serious games, or gamification, with 44 referring to their artifact as a game and 16 prioritizing the word gamification. On the other hand, 30 % referred to systems using terms such as labs, artistic practices, workshops, experiments, storytelling activities, gardens, or playful interventions.

36 % of the outputs described a fully analog artifact, 36 % a digital one, and the remaining 28 % were either hybrid or a combination of separate analog and digital systems. Most outputs, 91 %, described multiplayer interventions. In most cases, their duration exceeded one hour and often extended to multiple weeks and months (Table 2).

Beyond their own duration, some interventions integrated play and games with larger systems, e.g., teaching, training, or participatory processes. 40 % of the outputs did so clearly but most gamified their entire process, whether short and simple or long and complex. Where games were integrated, two salient types of intervention were observed: those integrating games and play with lectures (e.g., A06, A31, A50) and those using games and play as part of larger creative activities, such as workshops (e.g., A13, A15, A37, A52).

In terms of their real-world connections, 51 % described systems involving real locations with personal implications for the players, while 27 % simulated them via methods such as game boards or illustrations representing possible futures. Meanwhile, 22 % did not involve real locations or did not describe the system in enough detail.

Furthermore, 63 % of the studies mentioned direct dependencies and embeddedness in pre-existing digital sustainability technologies, including apps, monitoring devices, and renewable energy production systems with possible integration capabilities.

3.3.2. *Play and gameful design features*

"A game is a rule-based system with a variable and quantifiable outcome, where different outcomes are assigned different values,

Table 2
Number and percentage of outputs by gamified intervention duration.

Duration	Output number and percentage
One or more years	14 (16 %)
Weeks to months	15 (17 %)
Hours to days	31 (36 %)
One hour or under	7 (8 %)
NA	19 (22 %)

the player exerts effort in order to influence the outcome, the player feels emotionally attached to the outcome, and the consequences of the activity are negotiable." (Juul, 2005, p. 36) While this definition does not apply to all systems adding a playful layer to activities typically framed otherwise (e.g., not all outcomes may have clear values assigned, as in "winning" or "losing" the game), the core idea of a rule-based system in which players exert effort to attain an outcome is valid for all interventions analyzed here.

Thus, we start by classifying their goals—What did players try to achieve? What objectives did they fulfill by acting in the context of the intervention? Three categories can be listed. First, 42 % of outputs invited players to participate in a simulation or learning challenge with defined goals. These were typically termed "games" or "serious games" although this category also included evaluation studies with gamified digital systems (A25, A29), playful visualizations (A45, A60, A68), and field trips (A81). Second, 38 % described design, creation, and data contribution processes as the system's central activity and aim. Finally, 20 % included systems where participants engage in direct consumer action. These interventions typically used the term "gamification."

While the goal types above imply different levels of open-endedness, processes leading to similar objectives can still allow for varying degrees of participant autonomy. Of the 36 outputs with defined goals based on simulations or learning challenges, 18 focused on top-down education and learning, 15 left more space for open participation and autonomous decision-making, and three aimed to direct behavior change (A01, A25, A29). Of the 33 studies in which participants aimed to design and create, one (A12) did not grant extensive autonomy and had players designing in the confines of a computer simulation, learning about adaptation but with their creations not directly influencing adaptation infrastructure policy. Finally, of the 17 outputs in which players focused on direct consumer action, not all prioritized directed behavior change—two focused on top-down learning and skill acquisition as the main conduit to behavior change (A07, A61), while two presented behavior change apps co-created with citizens (A48) or students (A62), thus prioritizing participant autonomy at one point in the process.

In terms of game elements (Yee, 2006), all outputs but one (A45) included achievement-based elements, at least a challenge in one way or another. Of these, 26 % had in-game rewards for players and 39 % included some form of competitive multiplayer challenge. Meanwhile, 65 outputs (76 %) included social elements, which promote player socializing, relationship development, or teamwork; of these, 92 % involved some form of collaboration. Twenty-four outputs (28 %) included both competition and collaboration. Regarding game elements that support immersion, such as avatars, characters, stories, and virtual words or maps, 49 outputs (57 %) mentioned at least one. Finally, salient resource types reported across the sample include facilitation and/or debrief (41 %), real-world rewards (15 %), and internet of things (IoT) technology (9 %).

3.3.3. Design aspects for innovation and change

The interventions designed reached different levels of innovation (Ceschin and Gaziulusoy, 2019). Three focused on product innovation, meaning on a single product's life cycle and user-product interactions (A40, A66, A69). 21 % focused on the next level, the product-service system, which refers to combinations of products and services to impact larger consumption habits. 24 % focused on the spatio-social level, which refers to community practices. Finally, 51 % described systems targeting socio-technical level impact, that is, aiming to fulfill societal needs such as energy provision, nutrition, and mobility in a radically different way.

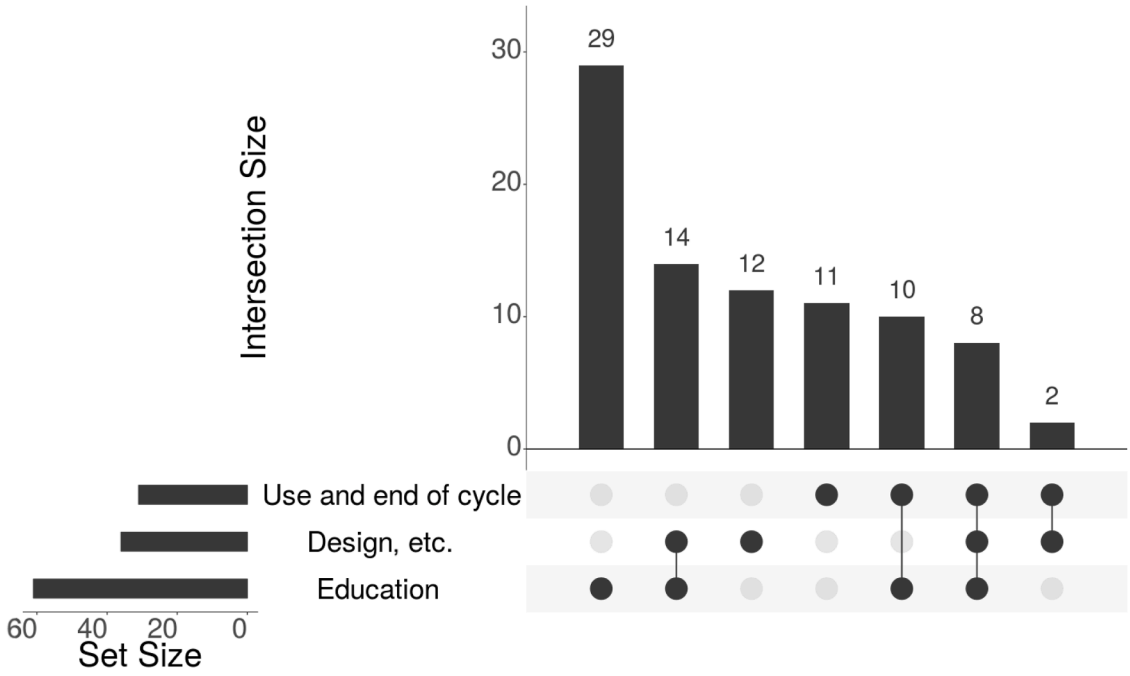


Fig. 6. Number of outputs by change type, including (a) use and end-of-cycle, (b) design, policymaking, production, and/or organizational change, and (c) education.

Changes were also promoted across different areas of human practice (Guillén et al., 2023). 71 % of the outputs focused on education; 42 % aimed to alter how design, policymaking, production, and/or organizational change occur; and 36 % included impacts on use and, on occasion, end-of-cycle (see Fig. 6).

Finally, we examined whether the interventions focused on practices, i.e., everyday activities, in a way that (a) considered the actors and the environment, (b) did not focus only on individual agency or systems, and (c) integrated within the process to be shaped. Most outputs described interventions that were not particularly practice-focused—instead, they related to pro-environmental behaviors without identifying clear practices, incentivized behavior change without technical assistance or particular regard for the activity's environment, centered on provision systems, or proposed simulations or creative practices outside of everyday life. However, 27 % included a degree of practice orientation, i.e., collective gardening, eating, mobility, and energy-consuming practices aided by technical implementations or meaningful guidance. Here, gamified interventions aiming to transform everyday professional practices, even if situated in real projects, have not been considered as practice-focused, since they transform, e.g., sustainability-focused educational and communication processes, but they do not transform quotidian day-to-day sustainable practices themselves.

3.4. Research characteristics

RQ3 aims to understand the research methods used in the studies. Here, we describe the sample sizes, data collection and analysis methods, and whether comparisons at different times or between designs were made. Sample sizes ranged from three to several thousands, with the most common range being under 50 participants (see Table 3). The supplemental file contains precise numbers where available.

The data collection methods reported were questionnaires (50 %); observations (42 %), including recordings, notes, and non-systematic data logging by researchers; data logs and other participant-generated outputs (40 %), interviews (20 %), debriefing, focus groups or panel discussions (20 %), and documentation sources (5 %). Eleven studies (13 %) did not specify their data collection methods. The collected data were analyzed in qualitative and quantitative ways (see Fig. 7), but the method was not clearly reported in eight cases (9 %).

73 % of the outputs did not use any form of comparison, 17 % employed pre-post measurements, and 9 % used other conditions, reference cases, or data suitable for comparison. These comparisons involved immersive virtual reality gaming versus traditional instruction (A69), gamified versus non-gamified platforms (A22, A75), temperature visualization using human characters versus abstract visuals (A60), a control city versus one in which a cycling initiative was implemented (A30), a fictional reference case (A44), consumption during hours with high renewable energy generation versus hours with low renewable energy available (A39), and a control created by using statistical matching techniques and data from previous years (A43).

3.5. Outcomes

We classified outcomes relevant to sustainability transitions as psychological results—when they refer to cognitive or affective changes—and behavioral results. This operationalization of engagement as a cognitive, affective, and behavioral state of connection with an issue has been widely used in the promotion and analysis of engagement with sustainability topics (Lorenzoni et al., 2007). While the first two refer to knowing, feeling, caring, and being motivated, and are thus labelled as psychological here since they refer to internal mental states, the third describes observable action. Since gamification targets both mental and behavioral outcomes as well (Hamari, 2019), this classification has been used in gamified sustainability engagement before (Fernández Galeote et al., 2021). This section also provides an analysis of the learning interventions' reach in terms of learning loops as well as the role of player experiences in the outcomes reported. The supplemental file contains more detailed information on all these variables.

3.5.1. Psychological results

71 % of the outputs reported a result connected with participant learning, i.e., increased awareness, acquired knowledge, improved skills, or reflection. Thirteen papers included implications and caveats associated with mixed results or lack thereof, which may be useful for future designers and researchers. They are summarized in Table 4.

Table 5 shows the classification of the 61 learning results, plus our understanding of potential learning from the rest in parentheses. Individual and group-level single loop learning results were the only ones resulting from the use of inferential analysis methods.

Meanwhile, 44 % of the outputs reported affective outcomes related to attitude, concern, self-efficacy, or a sense of connection with or interest in the topic. Four included clear caveats worth highlighting. In A08, the intervention had only a slight impact on attitudes and social norms, and its effects on intentions were mixed, specifically in terms of intending to commute actively but not by public

Table 3
Number and percentage of outputs by sample size.

Sample size	Output number and percentage
3–50	37 (43 %)
65–100	11 (13 %)
117–375	11 (13 %)
1200–92,500	4 (5 %)
NA	23 (27 %)

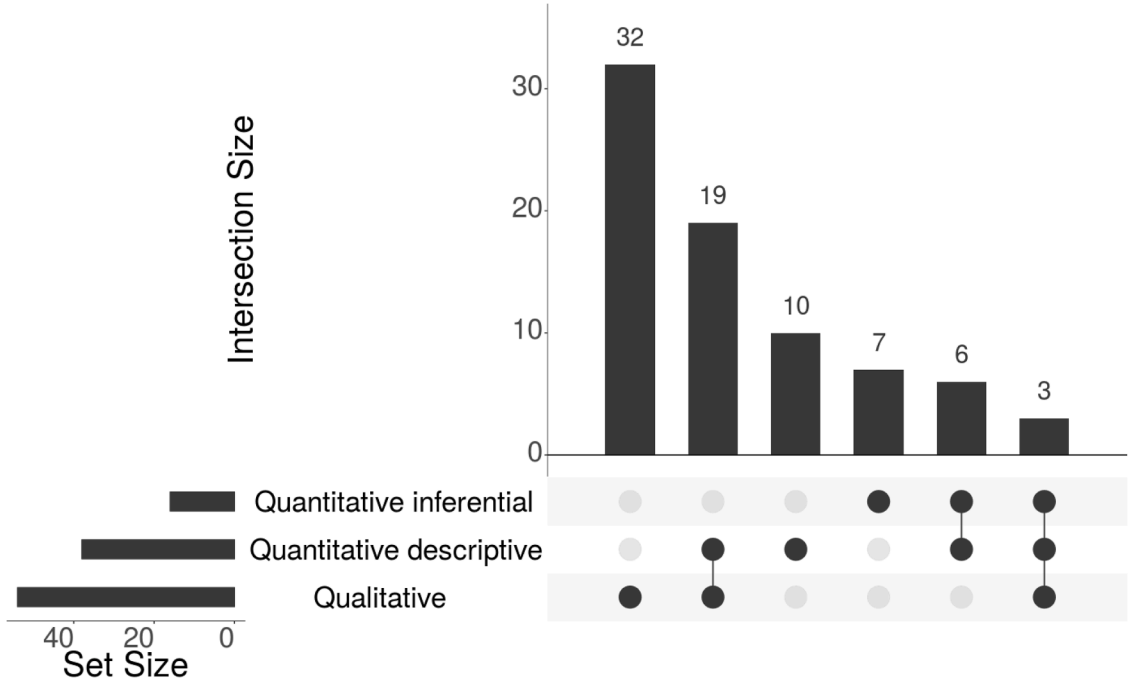


Fig. 7. Number of outputs by data analysis methods used.

Table 4
Caveats and implications of learning outcomes.

Output ID	Caveat summary
A01	A time limit can increase enjoyment and a sense of competition, but it can also reduce the time devoted to discussion and the content experienced.
A09	Knowledgeable stakeholders may notice technocratic perspectives and wish for a more socio-technical orientation instead.
A18	Limited time and not complementing the game experience with, e.g., written information may explain comparatively low self-reported learning.
A24, A28	Multiplayer serious games may not be sufficient to promote the deeper forms of learning needed for sustainability and energy transitions (e.g., anticipatory triple loop learning). These require intentional design, facilitation, and debriefing.
A26	Simulation games for professionals can be useful boundary objects for learning, but they may not be able to answer complex questions themselves.
A32	Serious simulation games may bring a limited increase in understanding and interest in the topic, given individual player differences.
A45	This article did not report any clear learning outcomes. Due to contextual factors such as placement and functional factors like screen size and interaction possibilities, most students reported that they did not learn anything from a playful visualization about solar energy.
A47	Disruptive players in an academic setting can affect others' learning attainment in games.
A51	Even after completing a behavioral intervention, many participants may still be unclear about which behaviors are sustainable.
A59	Excessively difficult challenges can lead to student demotivation, and insufficient support (e.g., not enough meetings with expert stakeholders) can affect the quality of their learning.
A63	Excessive complexity can limit engagement and reflection.
A85	Excessive challenge difficulty coupled with a lack of guidance, time limitations, and broken elements can lead to a negative experience and limited learning.

Table 5
Observed (and potential) learning outcomes according to social interaction level (vertical) and learning forms (horizontal), in number of outputs.

	Single loop	Double loop	Triple loop	Total
Societal	0 (+3)	2 (+1)	6 (+4)	8 (+8)
Group	20 (+7)	18	10 (+2)	48 (+9)
Individual	5 (+8)	0	0	5 (+8)
Total	25 (+18)	20 (+1)	16 (+6)	61 (+25)

transport. Here, participants found some generic behavioral suggestions inconvenient. In A13, there was interest in diversifying agricultural systems and produce, but a lack of labor availability was blamed for a lack of interest in developing organic coffee production; therefore, demographic changes can impose limitations that are difficult to overcome through personal engagement alone. In A22, ceiling effects in attitude may have limited the effectiveness of the intervention, where non-significant differences in attitude

and engagement increase were reported. Finally, A34 reported no effect on willingness to change energy-saving and efficiency attitudes and behaviors, which was attributed to a need for more guidance and preexisting engagement.

3.5.2. Behavioral results

Most outputs, 81 %, reported some form of behavioral result. Of the 110 unique behavioral results inductively classified, most had a direct real-world impact (see Table 6), while the rest referred to behaviors that may be relevant to sustainability but were confined to the game.

Of the outputs reporting behavioral outcomes, 12 indicated some caveats associated with contextual or design limitations (Table 7).

3.5.3. Mediating experience and design features

Experiential outcomes were reported in 60 % of the outputs. They included forms of hedonic value (24 % of the total sample) such as enjoyment, fun, or immersion; forms of pragmatic value (31 %) including preference over other methods, satisfaction, and recommendation; and motivation (16 %). The details of all these experiential outcomes can be found in the supplemental file. In addition, negative experiences were reported in 31 % of the studies. These described challenges in engaging with gamification, including initial reluctance to participate (A10) due to issues such as skepticism (A65) and distrust in outsiders (A35). A difficult start can predispose participants negatively for the rest of the experience (A19). In some cases, a lack of previous experience or excessive difficulty can negatively affect engagement (A17, A59, A63, A85) and derived outcomes. Players tend not to appreciate challenges that bear no personal relevance or where the effort seems to outweigh the benefit (A33)—a so-called "unpleasant sacrifice" (A51). Because players exhibit a variety of preferences and interests, games may have their effectiveness limited in some cases (A32). Similarly, selecting the wrong context (time, place, or both) can also severely limit engagement and thus other sought outcomes (A45). However, games are sometimes considered just not entertaining enough (A42). Lack of trust among players, for instance when they are industry actors who compete to some extent (A27); the presence of disruptive players such as noisy students (A47); and conflict or disagreement (A58, A74) can also negatively impact engagement. Finally, technical difficulties were reported in some cases (A16, A76, A85). Beyond gamified systems, researchers may also encounter problems related to control conditions not being attractive enough and being abandoned by participants (A22).

65 % of the outputs included some information on system elements, activities, and context characteristics leading to outcomes relevant to learning, affect, behavior, or system engagement and performance. The data matrix in the supplemental file includes a complete list of contextual and design aspects and their reported effects, either positive, negative, or mixed. Aside from the caveats to learning, affective impact, and behavior discussed in the preceding subsections, notable positive examples include facilitation (A12, A74) and metaphors (A14) reinforcing learning; playful and non-judgmental environments encouraging trust and connection between participants (A37); and emotionally involving tasks providing a sense of meaning and motivation (A11).

4. Discussion and agenda

This section summarizes our findings relevant to the RQs and outlines three agendas for future research and practice derived from them—a contextual one (RQ1), one for design and implementation (RQ2), and one focused on empirical research (RQ3 and RQ4). Given the variety of interventions reviewed across these foci of interest, we anticipate that the following recommendations will be helpful to researchers, designers, and educators interested in gamified interventions for sustainability education, affect, and behavior change at large, but may also prompt them to consider wider aspects in terms of what forms playful designs can adopt, and how to integrate sustainability changes into larger transition dynamics.

4.1. Contextual agenda

Our contextual agenda relates to the analyzed studies' approach and fit with their targeted transitions. Here, we discuss and propose ways forward in relation to gaps regarding conceptual positioning within sustainability science (point 1); targeted actors and sectors (2); how said actors are engaged (3); consideration of technical and ecological systems in transitions (4); and geographical location (5) and scope (6).

First, almost a quarter of the sample lacked explicit definitions of sustainability—some gamified STs were justified through the need to act against climate change (A59) or reduce an ecological footprint (A48), while others focused on the importance of activities such as

Table 6
Reported behavior types.

	Direct real-world impact reported	No clear direct real-world impact reported, even if potential
Consumer behavior	17	0
Participation in activities and behaviors other than consumption	17	0
Social interaction	28	14
Artifact and data contributions	27	5
Other later influences reported	2	0
Total	91 (51 outputs)	19 (19 outputs)

Table 7
Caveats and implications of behavioral outcomes.

Output ID	Caveat summary
A07	Behavior alone may pale in comparison with the role that technical improvements can have in reducing energy consumption.
A08	Participants found some generic behavioral suggestions inconvenient.
A17	A free-form simulation with many affordances and imprecise goal- and communication-oriented features can lead to disorganization. Then, a lack of participants where researchers want them to interact can have a detrimental effect on data generation.
A21	Lack of incentives can lead to low citizen participation.
A30	The behavior changes brought about by gamification may not be sustained in the long term, especially if the incentive structures created are discontinued.
A33	Design limitations can allow for spurious effects, for example registering sustainable commutes that did not substitute unsustainable ones.
A35	Top-down initiatives can entail exclusionary behavior by participants and rejection by outsiders; citizens may have a perception of imposition and refuse innovations, preferring to keep their autonomy.
A39	Day-to-day habits may not fit renewable energy production times, severely limiting the potential of behavior change interventions.
A43	The behavior changes brought about by gamification may not be sustained in the long term, especially if the incentive structures created are discontinued. Some goals can be achieved through efficiency or sufficiency without observers clearly knowing which one is being used.
A51	Behaviors that are costly in terms of time or money are less likely to be implemented.
A57	Mobility behavior change may need stronger incentives than, e.g., non-tangible rewards and the possibility of sharing information and experiences.
A59	Excessively difficult challenges can lead to student demotivation, and insufficient support (e.g., not enough meetings with expert stakeholders) can affect the quality of their outputs.

recycling (A66) or promoting renewable energy (A09). The lack of further anchoring indicates that sustainability often remains undertheorized and insufficient for a critical approach to ST gamification. Outside of this subset, institutional frameworks were abundant, ranging from the international (e.g., the Paris Agreement, A56; United Nations, A86) to the regional (e.g., the European Union, A27) and the national (e.g., German legislation and policy, A77). In contrast, critical approaches such as ecofeminist, indigenous, and post-extractivist approaches (A67), or more-than-human (A54), were comparatively scarce.

ST research is a complex field of study open to multiple theories and interpretative frames (Truffer et al., 2022). Yet, it necessitates theoretical anchoring in well-defined ontologies and epistemologies (Geels, 2022), as well as clear frameworks for action. In particular, critical perspectives have gained salience in ST research (Truffer et al., 2022)—from issues of power and justice (Avelino et al., 2024; Köhler et al., 2019) to the need to critically engage with capitalism to assess STs' sustainability, adopt STs beyond the Global North, and explore more diverse transition pathways (Feola, 2020). In line with these calls for more criticality, we found that most outputs with a critical approach to sustainability were published in 2024 alone, including perspectives on environmental justice and decolonization (A64), degrowth (A71), and social justice in the Global South (A70).

While it may be argued that small-scale interventions do not require lengthy contextualization, neglecting an intervention's positioning in relation to STs does not imply that it lacks one—rather, it lacks transparency about the assumptions that have been made. Given that social, political, and economic paradigms such as capitalism define the shape of the socio-technical systems within them, ignoring our own theoretical standpoints and societal roles as researchers limits "the capacity of STR to support future societal sustainability transitions" (Feola, 2020, p. 242). Gamified ST research requires critical engagement with sustainability-relevant theoretical and practical concepts to adequately contextualize and assess findings and their implications.

Thus, we formulate our **contextual agenda point 1: Gamified ST studies should critically engage with well-defined and diverse sustainability paradigms.**

Second, we found a lack of interventions focused on industries and businesses beyond local energy, with public and private practitioners being rarely targeted as direct participants. However, interesting examples emerged, including a gamified system for multi-level cooperation in food transitions (A27), a game-based trade-off charting tool for livestock transformation (A26), and a role-playing game designed to engage diverse stakeholders in multi-criteria decision-making for electricity grid reinforcement (A46). While STs focusing on adaptation to environmental impacts were rare and exclusively centered on urban environments (A17, A65), one of them involved stakeholders across the board (A65). These examples suggest that gamification can be utilized to engage practitioners in exercising their various forms of power to facilitate transitions.

Future studies are thus encouraged to "systematically and explicitly analyse the complex diversity of the (ascribed/constructed) roles of different sectors and actors in multi-level transition dynamics" (Avelino and Wittmayer, 2016), p. 638) to appraise where and how gamification can be introduced most effectively. In other words, there is a need to involve practitioners through mechanisms informed by a multi-actor perspective that considers their sector, level, role, and power relations within the ST (Avelino and Wittmayer, 2016). Furthermore, involving more public and private practitioners creates opportunities to effectively combine mitigation and adaptation perspectives in ST, as both political leadership and the public-private interface are important in stimulating resilient STs, i.e., able to withstand disturbances such as climate change impacts (Mendizabal et al., 2018). While desirable, involving practitioners in STs involves distinct challenges. On the one hand, they may be perceived as a threat, thereby destabilizing professional identity (Janssen et al., 2022). On the other, professionals may perceive gamification as unsuitable for serious contexts (Fernández Galeote et al., 2021) or insufficient to provide the level of trust needed between competitors (A27). Therefore, gamified ST interventions for practitioners must be particularly sensitive to these issues.

Here, we propose our **contextual agenda point 2: Industrial and governmental actors should be targeted more often, especially with a focus on promoting resilient sustainability transitions.**

Third, empirical research on ST gamification typically focuses on behavior and understanding as the primary outcomes to be

sought, including issues such as students' daily decisions (A16), energy community governance (A21), or engagement with scientific concepts (A11). Meanwhile, the attitudinal implications of directly engaging with STs such as school gardening (A41) or simulating energy transitions (A05) are much less prominent. Given that emotions are central to ST engagement (Martiskainen and Sovacool, 2021; Thelin, 2024; Valor et al., 2022), we must stress the capacity of play and games as tools for affective exploration and involvement (Hemenover and Bowman, 2018; Isbister, 2016).

Using gamification that considers and targets affect requires reconceptualizing ST processes as emotionally charged. It also requires theories of change in which gamification acts as an emotional pulling force capable of recruiting actors in STs beyond economic incentives—actor recruitment being a concern in socio-technical transitions and social practice (Laakso et al., 2021). While many strategies attempting to manipulate emotions toward sustainability have failed, effective techniques exist (Brosch and Steg, 2021). Gamified ST interventions should aim, first, to understand current roles, practices, and attached emotions, which will often require grounding from pro-environmental gamification at large (Fernández Galeote et al., 2021), given the relative lack of research focusing on emotions in ST (Martiskainen and Sovacool, 2021) and especially gamification therein, as per our findings. One caveat to consider is the fact that games, as other engaging media, often "manipulate player affect to create poignant experiences" (Lazzaro, 2009, p. 156). Thus, it is not only a problem that manipulating actors' emotions can have boomerang effects (Brosch and Steg, 2021); the ethical implications of said manipulations, even if they are successful and for virtuous ends, should be acknowledged and addressed.

From the above, we formulate our **contextual agenda point 3: Gamified ST interventions should consider the role of affect as part of engagement issues.**

Fourth, gamification primarily targets transformation at the socio-institutional level, whether at the community level (A02) or by targeting individualized practices within (A04). In contrast, only a minority of interventions focus on socio-ecological issues, such as community gardening (A10) or agroforestry (A13), and just over half clearly support a direct socio-technical transition, for example, involving energy system software (A63) or data-driven agriculture (A82). A desirable future development consists in using gamification as a means to support transitions beyond social change alone; for example, by targeting situated practices through implementation within sustainable provision systems (A58), experimenting with alternative technological configurations such as solar-powered and anti-obsolescence entertainment (A83), or creating artistic nature-based solutions that combine socio-technical, socio-institutional, and socio-ecological transitions (A53).

Here, we join calls for designers' direct involvement with pre-existing socio-technical and socio-ecological niches from various related areas of research, including sustainable human-computer interaction (Ganglbauer et al., 2013) and design for STs (Gaziulusoy and Erdoğan Öztekin, 2019). Herein lies potential to draw from more-than-human design or posthuman design, which emphasize the need to see design within a broader set of more-than-human relations and values and have been applied in playful and artistic methods for sustainability (Genç et al., 2024; Tomico et al., 2024). More-than-human approaches have been employed in the analyzed outputs to question why so-called invasive species are targeted for elimination and propose instead a playful relationship with them (A54) and to visualize and discuss future food-sharing initiatives (A68). While involving technical and ecological systems and their actors may bring additional complexities and resistance, our call for a more critical and integrated use of gamification for ST requires it.

Therefore, we propose our **contextual agenda point 4: Gamification should aim to support socio-technical and socio-ecological transitions directly.**

Fifth, Western Europe and the US concentrate an overwhelming majority of the outputs. Energy and mobility planning in India (A20), integrated crop-livestock systems in Brazil and Paraguay (A52), sustainable livestock transformation in various African countries (A26), and ecotourism behavior among Chinese students in Malaysia (A75). These interventions align with identified transition needs in context, especially where incumbent systems are different, and show that gamification can be implemented as a companion tool. Yet, the fact that these interventions occurred in non-Western contexts does not necessarily imply that non-Western perspectives are being explicitly embraced, as has been called for in ST research at large (Feola, 2020). Two positive examples are A70, which rejects techno-economic thinking that prioritizes large-scale fisheries in favor of situated and socially sensitive thinking in Uruguayan small-scale fisheries, and A67, which explores Latin American art practices that defy "Western-centered views on energy and energy infrastructure" (p. 1).

These examples can be seen as part of a decolonial turn (Ghosh and Mguni, 2025) in ST research, which aims to understand and act more in the Global South, where STs remain underexplored despite exhibiting unique forms of innovation (Oates, 2021). A challenge is the fact that transitions present similarities and differences across geographical contexts, including different priorities, material conditions, and conceptual assumptions that can defy theoretical assumptions made in the West (Köhler et al., 2019). Local knowledge and direct participation are crucial in avoiding external impositions. Where international cooperation between institutions from the South and North are advantageous (e.g., A13, A26, A52, A67, A70), they may develop this research area by combining their strengths but minding embedded power asymmetries toward equitable engagement (Flint et al., 2022; Sami, 2023).

Based on the above, we articulate our **contextual agenda point 5: Gamified ST should further extend its research beyond the West.**

Sixth, and continuing with geographical matters, most gamified systems involved or simulated real locations, as expected given that STs are context-sensitive; however, the majority remained at the local level or below. From a multi-level perspective (Geels, 2002), gamification seems to be implemented at the micro level as novel configurations that may influence the patchwork of regimes, but lack larger integration or systematicity. Thus, ongoing cultural, landscape-level shifts towards societal gamification and sustainability do not seem to clearly intersect at the *meso* level. In other words, the gamified systems reviewed here target individual and even community agency as part of a broad cultural shift (Hamari, 2019), but the degree to which this has resulted in sustainable gamification having a structural impact (Archer, 2020) is unclear.

STs target a variety of large-scale systems across sectors (e.g., energy, transport, land use and agriculture) that are maintained and

reinforced at the supranational level; indeed, STs themselves are governed through a myriad of transnational goals, agreements, and sector-wide standards that permeate different levels (Manning and Reinecke, 2016). For gamified STs to increase their impact, a desirable future involves scaling up these interventions and research. This may occur through the creation and expansion of multi-local interventions (e.g., A19, A26, A67, A71) and through regional and international-scale gamified projects, which have resulted in global collaboration between scientists, artists, and practitioners (e.g., A11, A19, A37) and stakeholder engagement with large-scale energy projects (e.g., A18, A46).

On the one hand, to support replication and reuse, gamified ST interventions can include open access to systems and data where allowed, and recommendations on how to use and modify them. On the other, research should explore how gamified ST interventions can be conducted at the regional and international level, including possible resource limitations and tensions between scale- and context-agnostic systems (A19) and targeted environments. This will occur in a context in which complexities such as professional identities (Horsbøl, 2018) and multi-level governance (Ehnert et al., 2018) present challenges common to any ST process, but also one where play is ubiquitous but far from being used for the common good at a large scale (Harviainen et al., 2025). Again, taking into account contextual differences and adopting sound strategies to upscale ST engagement technologies (Wemyss et al., 2023) will be crucial.

From the gaps and examples presented, we derive our **contextual agenda point 6: Gamification should be applied and expanded regionally and internationally.**

The six points above seek to renew the contextual understanding and fit of gamified ST interventions and research, often in line with current broader ST research concerns. By advocating for a more explicit and critical integration of conceptual bases (1), this agenda aims to clarify scientific communication, deepen theoretical reflection, and ground practical advancement. By calling for including practitioners capable of implementing resilient STs (2) and considering actors' affective and attitudinal engagement (3), it aims to deepen our understanding of participants and their relationship to future interventions. Finally, by embedding this area of practice and research in socio-technical and socio-ecological transitions directly (4) and implementing it further beyond the West (5) and at larger scales (6), its goal is to increase the impact of gamification across systems and geographical contexts.

4.2. Design and implementation agenda

Our design and implementation agenda relates to how the interventions have been conceptualized, developed, and applied to their various contexts and STs. Here, we discuss and propose ways forward based on observed gaps regarding effective design strategies (1), catering for the interests of individual players (2), and justifying player constraints and agency (3). We also advance different design perspectives regarding disciplinary understandings of games (4) and tackle goal diversity (5) and the types of elements used (6). Finally, the agenda touches upon the technological underpinnings of gamification, including engaging technologies (7) and other tools (8).

First, almost a third of the outputs reviewed explicitly reported negative player experiences. Some examples included lack of context fit (A45), excessive difficulty (e.g., A17), insufficient relevance (e.g., A33), and lack of flexibility to accommodate diverse interests (A32). This echoes previous findings in which some gamified pro-environmental interventions were perceived as "unfit for professional settings, too difficult or complex, or disconnected from [the players'] interests" (Fernández Galeote et al., 2021, p. 22). However, if a primary advantage of gamified interventions is their wide appeal and motivational pull (Hamari, 2019), designers should ensure that their artifacts exhibit these characteristics for as many of their players as possible, as strongly as possible. Based on the issues observed, some of the most critical measures include providing tools that directly support capacity to perform ("I can do it"), ensure personal and contextual relevance and appeal ("I want to do it"), and foster trust among players, the organizers, and the system ("this context is helping me do it").

One important point regarding the issues above is that they can be addressed in advance—initial reluctance can turn into participation when the allure of novelty or clear purpose manifests (A10), difficulty can be tackled through scaffolding, and individual differences (A32) or the specific needs of defined audiences (A27, A35) can be identified through user research. Hence, relevant stakeholders should be involved in the design process (A01, A28, A32, A38, A48, A52, A62) as part of a user-centered design process in which designers can learn about the players' skills and predispositions towards games, the topic at hand, and other relevant factors (Fernández Galeote et al., 2023b). Furthermore, a user-centered mindset can be combined with existing design frameworks specifically created for STs, such as the future-oriented transition design approach by Irwin (2015) or the design science research approach for transition studies by Wiegmann et al. (2023).

Our call for a more thorough ST intervention design process, i.e., one that acknowledges the complexity of context and the importance of understanding (often heterogeneous) stakeholders, mirrors calls for recognizing the complexity of STs in systems involving multi-agent interactions in policy-making (Mercure et al., 2016). While it is to be expected that design processes will be slower, more demanding in terms of resources, and prone to friction, considering the actors involved is the best way to craft an engaging and effective experience.

Hence, we propose our **design and implementation agenda point 1: More gamified ST design processes should consider effective (e.g., user-centered) strategies to promote engagement among diverse users.**

Second, almost all outputs were multiplayer, allowing players to collaborate or compete across a variety of settings and goals. This contrasts with environmental areas such as climate change engagement, where gamification has a more individualistic bend (Fernández Galeote et al., 2021). Yet, ST interventions may be informed by experiments in living, which modify "habits and habitats according to a fixed procedure" to find new ways to attain pro-environmental goals (Marres, 2012, p. 113) and may be conducted individually even if they are often shared with others. They resonate with play experiences where living and learning are personalized,

whether by design or by players taking advantage of design openness—e.g., by playing a video game in "vegan mode" (Westerlaken, 2017).

In addition to experimentation, ST interventions often include forms of individualized training and education (A42). In a conscious design effort, certain phases of instruction or engagement can use tailored gamification (Klock et al., 2020), interactive agent-based modeling, or AI integration (Bezzina and Dingli, 2023) to customize experiences when it is a priority or involving co-players is not possible. While no examples of these approaches were clearly found in the sample, A06 adopted an Afrofuturist approach to future visioning that is informed by the participants' personal background, resulting in deeply personal crafts and visions. In addition, several multiplayer interventions included customizable goals (e.g., A43, A62, A78, A81). A challenge implicit here is the fact that centering individual agency risks ignoring the limitations of personal behavior and opportunities in the wider context, such as forms of collective action (Ganglbauer et al., 2013). Thus, it is essential to promote practice-oriented interventions that support change in a situated manner, considering both individual actors and systems and environments.

Based on the above, we formulate our **design and implementation agenda point 2: Gamified ST interventions should involve more personalized practices and perspectives.**

Third, this review found that education was the most prominent type of change pursued toward supporting STs. Nonetheless, even learning outcomes were most often sought through open-endedness and player autonomy, rather than coaxing towards defined learning outcomes or persuading towards specific behavior change. Examples include A38, in which board game players collectively exploited and defined their own rules, and A23, which used a different board game to facilitate multi-stakeholder dialogue and collaboration. This suggests that similar goals can be achieved through various means.

Still, the reasons for choosing a system's degree of openness to player agency should be clearly defined and substantiated to identify best practices in relation to specific contexts, actors, and goals. A systematic gamification design framework (Morschheuser et al., 2018) can provide a clear design process structure based on engagement goals, while user-centeredness (Fernández Galeote et al., 2023b) can identify the opportunities and constraints of the actors and their context. Then, specific approaches to pursue engagement outcomes can be incorporated—some that were not seen or rare in this literature were behavior change techniques (Carey et al., 2019), instructional theory (Honebein and Reigeluth, 2021), or multimedia learning principles (Mayer, 2009) for targeted approaches, and unstructured play design (Smith et al., 2023) or speculative design (Chopra et al., 2022) to support more creativity-focused systems. All in all, researchers and designers should base gameplay on aim and context, supporting it with theory and evidence, including awareness of the limitations of their chosen approach (e.g., Brynjarsdottir et al., 2012; Wemyss et al., 2023).

These points inform our **design and implementation agenda point 3: Gamified ST interventions should intentionally define their degree of openness to player agency using specific knowledge, theories, and frameworks.**

Fourth, this review adopts an intentionally broad lens because playful and even game-like interventions may not always use terms such as game or gamification. Having opened our search and selection strategy to humorous, artistic, and playful activities more broadly, a third of the outputs found characterized playful or game-like interventions as labs (e.g., A49, A59, A65), artistic practices (e.g., A53, A84), workshops (e.g., A54, A83), experiments (e.g., A03, A50), gardens (e.g., A10, A54), etc. Having analyzed these outputs, we can identify a thread that weaves together playful and game-like interventions, regardless of the term used. Despite framing and design differences, they all seek to engage participants using playful methods; that is, they give engagement a front-and-center role and imprint an emotional quality to the topic at hand, no matter how real or relevant (Masek and Stenros, 2021).

Therefore, ST gamification designers should realize the many unconventional ways in which gamification exists, in the same way that researchers from areas such as human-computer interaction have called for open-ended systems and interventions that remain engaging without being necessarily called "games" or even "play" but instead simply "engaging" (Rogers, 2006) or "joyful" (Altarriba Bertran and Márquez Puig, 2024). Besides embracing more inclusive terms and conceptualizations, future research may expand the search even more towards potentially relevant interactive and participatory activities that prioritize engagement. A significant risk remains, though, in including loosely related interventions without a clear indication that they involve similarly engagement-prioritizing mechanisms. Our call to a pluralist approach to playful interventions and their syncretic design increases complexity and requires a sound understanding of how to design and evaluate unconventional engaging experiences, as well as clarity of explanation and design justification on the part of future authors.

As a result, we propose our **design and implementation agenda point 4: Design and research of gamified ST can take inspiration from engagement-prioritizing interventions beyond what has been traditionally termed gamification or games.**

Fifth, the outputs analyzed were divided into three categories—first, simulated scenarios such as energy board games (A18) and other learning challenges, such as a public engagement project that included a field trip (A81); second, design, creation, and data contribution experiences, including artistic interventions towards socio-ecological understanding and restoration (A50) and design challenges as part of formal education (A49); and third, systems for direct consumer behavior change, including for saving energy in buildings (A58), in transport (A30), and involving multiple habits (A51).

However, in the same way that games can be combined with other methods such as lectures, focus groups, and knowledge tests, the approaches above can be combined in fruitful ways, such as by integrating simulated scenarios for awareness-raising with action-oriented activities that connect general knowledge with situated action (e.g., Fernández Galeote et al., 2023a). This is an example of what can be achieved through multi-stage processes that focus on learning, creation, and behavior change at various stages. For example, a gamified intervention for actors to plan theories of change (A19) could be directly complemented by guided learning processes and follow-up support for implementation. In another case, a game was used as an entry point towards a larger process of farm coaching (A52). A possible risk of this approach is participant tiredness and loss of interest, which could be mitigated by deeply understanding the stakeholders' needs and preferences to provide a cognitively, affectively, and behaviorally engaging experience (Lorenzoni et al., 2007), including leveraging the social environment and activities outside of the game's frame. Similarly, longer

interventions would involve difficulties in measuring outcomes and their causes, e.g., when trying to isolate the effects of multi-step interventions. For each case, adequate data collection and analysis methods and epistemological points of departure should be sought (Geels, 2022).

Therefore, we articulate our **design and implementation agenda point 5: Different game and goal types can be combined within gamified ST interventions.**

Sixth, almost every system included achievement aspects, e.g., a challenge and performance feedback, and the prevalence of multiplayer systems was accompanied by a high presence of elements promoting social contact. Yet, elements promoting immersion were less numerous, even though a variety were found—e.g., roles to play (e.g., A01, A09), characters (A45, A79), visual worlds (A21, A33), and forms of storytelling, including through artistic participation (A6, A64). Combined with the need for further consideration of affect in STs, we believe that including elements aiming to immerse players in game worlds, their dwellers, and their stories, would increase the emotional impact of certain gamified systems (Hemenover and Bowman, 2018; Isbister, 2016) and could be explicitly leveraged to expose participants to different ways of knowing and living. For example, the aforementioned A64 describes storytelling used as a form of resistance to mining, and A17 uses a virtual world in which players can engage in ecosystem-based coastal adaptation.

Stories and their characters are some of the most commonly used elements in games to manipulate affect (Isbister, 2016). Besides the ethical concerns related to emotional manipulation mentioned in the contextual agenda, stories and characters are at risk of being perceived as stereotypical, patronizing, or disconnected from the stakeholders' lived experience, and thus of little or no value for them (A33, A51). In these cases, not only may immersion suffer, but also these stories' and characters' potential to spark reflection. To ensure that narratives remain engaging and effective, they should be not only deep and mature enough for the audience but also relevant to the context and actionable. Here, frameworks of use include Sheppard's principles for engaging climate change storytelling—tying it to the community, providing visuals, and connecting it to the big picture (Sheppard, 2012).

Thus, we offer our **design and implementation agenda point 6: Immersive elements should be used more in gamified STs to fully capture the appeal of games.**

Seventh, the use of digital media was observed in most interventions, including hybrid forms with face-to-face workshops allowing participants to discuss live while having computational support where it is important (e.g., simulation fidelity) (Fernández Galeote et al., 2021). While analog systems elicit the complexities of socio-institutional transitions and digital media can accurately show and mimic technical-procedural complexity, hybrid systems can capture technical-physical and social-political aspects inherent to socio-technical transitions accurately (see Mayer, 2009).

Yet, some technological affordances were rare, including engaging interactive technologies such as virtual reality (VR) (A49) and augmented reality (AR) (A60). While analog interventions facilitate social contact and emotional engagement with others and the world around us, immersive digital technologies can richly and visually depict futures and articulate technically complex simulations. As the studies in the sample that unequivocally used immersive VR and AR were limited to an educational design challenge (A49) and an interactive visualization for education about domestic temperature (A60), there do not seem to be many examples in the area of gamified STs. Nevertheless, examples of immersive technologies for sustainability awareness can be found elsewhere (Cosio et al., 2023), including gamified ones (Fernández Galeote et al., 2023a). Despite their advantages, these technologies have limitations—immersive VR is costly and requires technical expertise, although it may be suitable for use in learning situations (A49). Because VR can also be isolating if limited to single-user experiences, designers can explore social immersive technologies such as multi-user VR and AR.

Based on these observations, we propose our **design and implementation agenda point 7: Gamified ST interventions should explore the engaging potential of engaging technologies such as immersive VR and AR.**

Eighth, half of the outputs described innovations that proposed radical changes to societal need satisfaction; that is, beyond altering products and services, their consumption, or the community practices around them. Some examples include a collaborative backcasting game for local administrators (A12), a board game to elicit electricity siting stakeholder preferences (A18), a visualization of solar energy production for children (A45), and an IoT-powered tool to influence mobility behavior (A57). The contextual agenda already introduced how gamification should be used to support socio-technical transitions directly. Here, from the point of view of design and implementation, we recommend combining gamification with sustainability technologies and their associated practices to bring not just social innovation but also boost ongoing technical innovation, as design for ST should embed itself in pre-existing practices across levels rather than continuously create new niches and thus contribute to further atomizing STs (Gaziulusoy and Erdoğan Öztekin, 2019).

On the one hand, most of the outputs described clear dependencies on pre-existing sustainability technologies, such as IoT (A57). In essence, gamification is being developed in a landscape where digitalization, including big data analytics and the "smart" discourse (e.g., "smart cities"), is deeply affecting STs (Köhler et al., 2019). This suggests the usefulness of having repositories and libraries listing technologies that can easily and openly integrate with engaging methods and work to facilitate said integrations on both sides. One challenge here is technological obsolescence, a well-established phenomenon in digital gamification and serious games, which often lack sufficient long-term institutional support (Fernández Galeote and Hamari, 2021).

On the other hand, gamified interventions should be combined with other methods, as they can be more effective in this way in terms of, e.g., learning (Wouters et al., 2013). For example, in one article, success was attributed to combining first lectures with then games that exemplified them (A18), while another used a game as part of a larger coaching process (A52). In the same way that every instructional method has its place (Honebein and Reigeluth, 2021), designers need not be constrained to using only gamification—in STs, there are moments for listening, moments for practicing, and moments for direct transformation in which stakeholders will have little cognitive space for additional engagement. Games do not need to address everything.

In line with these observations, we formulate our **design and implementation agenda point 8: The fact that gamified ST is**

embedded in larger systems should be leveraged through the use of pre-existing technologies and complementary methods.

The eight points above seek to provide fruitful next steps in relation to gamification design and implementation in the area of ST research, by leveraging recommendations and principles from fields such as design, education, and human-computer interaction. This agenda aims to increase the effectiveness of designs by calling for effective user-centered design strategies (1), taking into account the needs of individual players (2), and a stronger justification for designed player constraints and agency (3). It also aims to increase design diversity by expanding the current array of artifacts beyond traditional games (4) and advocating for combining diverse goals (5), incorporating immersive elements (6), and integrating immersive technologies (7). Finally, to foster both effectiveness and variety, the agenda highlights the need to leverage pre-existing tools and techniques alongside gamified systems (8).

4.3. Empirical research agenda

Our empirical research agenda relates to the studies' characteristics as research artifacts and focuses on scientific process and pursued outcomes, rather than contextual or design-specific issues. Here, we highlight and contextualize gaps and opportunities in relation to study size and duration (point 1), study designs (2), data collection and analysis methods (3), critical appraisal of results (4), and outcomes of interest (5) and their antecedents (6).

First, sample sizes under 50 participants were common, and unclarity in numbers was observed in many cases. Most of these involved qualitative case studies in which a strict participant count was not possible (e.g., A27, A35) or where the data were collected and analyzed based on households rather than individuals (A43). In another case, quantitative data was extracted based on game sessions, but the number of participants was not mentioned (A20). Furthermore, few outputs described interventions spanning a year or more, which paints a picture in which large-scale studies, both in terms of sample size and timespan, were not the norm—suggesting, again, that the typical gamified ST intervention is rather limited in its reach. This aligns with the broader design landscape for STs, where long-term and open-ended interventions are not common—a situation that has prompted a call for greater ambition in this regard to understand the potential roles these interventions could play beyond pilot experiments (Gaziulusoy and Erdoğan Öztekin, 2019).

In terms of sample size, we suggest that gamified interventions involving large groups should document the participants' demographics and relevant background variables, as well as their forms of involvement. An example of a large-scale intervention that combines a defined audience with precise quantitative data is A30, in which a cycling promotion campaign targeting schoolchildren and their families was undertaken in Poland. Thus, a well-justified and sufficiently large sample can provide rare insights on the effects of gamification at the scale of, e.g., a local school system. Of course, the adequacy of a sample size depends on the research question: Inquiries that are best answered through qualitative approaches with small sample sizes remain important in ST research (e.g., Marres, 2012), as small-scale experiments can inform visions of novel systems (Gaziulusoy and Erdoğan Öztekin, 2019). In our analysis, we found that they can provide a deep understanding of playful experiences and affect (e.g., A06, A14, A28, A37, A41).

In terms of duration, longitudinal interventions and studies remain important to understand the impact of long-term gamification systems (e.g., A30, A50, A58) or their delayed effects (A43). These studies are challenging, given the need for long-term funding and the requirement to keep participants engaged throughout the process. Even in that case, longer-term impacts may be out of scope, since many transitions take years and decades, thus surpassing the duration of most research projects. However, a long-term view is characteristic of both STs and ST research (Köhler et al., 2019; Zolfagharian et al., 2019), and evaluation frameworks have been proposed for the appraisal of future ST impacts (Williams and Robinson, 2020).

Based on our findings on sample size and study duration, we propose our **empirical research agenda point 1: Gamified ST research should clarify and justify its sample sizes and conduct larger, longer studies to increase its impact.**

Second, a quarter of the outputs analyzed included some form of comparison, most of which were pre-post. This means that questions related to gamification's effectiveness are being asked, but typically, there is no explicit benchmark against which to compare the results. Some examples of pre-post comparisons include A05, which reported increases in energy transition knowledge and self-efficacy among students; and A31, A42, and A85, all of which also reported increases in student knowledge. While these results imply that gamification was effective, there was no alternative method to compare it with.

Given that many gamified interventions appear to remain at the pilot experiment stage, several of the analyzed articles were able to compare effectiveness with a different condition. Although not demonstrated in any study here, future gamification experiments can incorporate value-added research (Mayer, 2019; Buchner and Kerres, 2023) and other approaches to investigate how participant characteristics impact the results. However, in the ST field, controlled experimental designs are often seen as undesirable given their lack of similarity with messy, embedded practices and contexts (Kaljonen et al., 2019). Despite this limitation, the sample analyzed included field experiments about transport and energy in which it was possible to use a different city as control (A30), a fictional reference case (A44), consumption during hours with a different energy mix (A39), and data from the years before the intervention (A43). Therefore, comparisons can be made even in relatively messy environments, and comparative studies remain a possibility for future gamification research (Köhler et al., 2019).

From the above, we formulate our **empirical research agenda point 2: Control groups and comparisons should be used where appropriate, including in real-world settings.**

Third, the sample analyzed showed a variety of data collection and analysis methods, with mixed methods being used in a quarter of the studies. Given that relevant gamification questions range from effectiveness in cognition, affect, and behavior to individual psychological experiences, a mix of methods is often appropriate. For example, A60 combined qualitative data with inferential statistics to capture both skill improvements and the motivational and engaging effects of visualization in an AR educational game; A85 used similar methods to assess learning and acquire a nuanced understanding of the player experience in an energy transition-themed

escape room; and A86 used mixed methods to capture student perspectives on pro-environmental behavior, their shifts in sustainable consumer and social action, and their experience with the gamified system. These examples highlight the importance of using mixed methods where adequate, in line with calls in ST research at large to combine qualitative case studies, which remain a majority in the field (Hansmeier et al., 2021), with quantitative and mixed methods approaches as adequate to the research question (Truffer et al., 2022).

Therefore, we propose our **empirical research agenda point 3: Mixed methods and methodological variety should be used in gamified ST research to clearly elucidate cognitive, affective, behavioral, and playful experience outcomes.**

Fourth, caveats were observed in multiple learning, affect, and behavior results, although none of the articles reported a clearly negative outcome from using gamification. Clear reporting of an intervention's limitations in impact and explanations for why effects may have differed from anticipated, as collected in Section 3.5, will be of great help to those seeking to advance the area of gamified ST engagement.

STs involve complex socio-technical realities that are not easily disentangled. Gamified ST research should collect and analyze data to indicate "how outcomes are generated by actors and causal mechanisms in structural contexts" (Geels, 2022, p. 4) through, e.g., an agentic-processual approach to critical realism (Geels, 2022). While small-scale or controlled gamified ST interventions may not contain in themselves the degree of causal complexity presupposed to STs themselves (i.e., conjunctural, configurational, event-chain; Geels, 2022), their embeddedness amid much more complex processes and contexts suggests the need to adopt adequate heuristic devices for causal explanation (Geels, 2022). Additionally, the use of analytical frameworks to validate the sustainability of processes and their impacts is recommended (Williams and Robinson, 2020).

In this review, we have seen how understanding contextual factors remains essential to adequately evaluate the impact of gamification. The studies analyzed provide valuable insight on how, e.g., experts demand game models beyond technocratic approaches (A09), deeper forms of learning often require complementary methods (A24, A26, A28), demographic trends can impact behavioral change (A13), costs affect behavior change (A51), and the role of technical improvements on energy consumption versus behavior change alone (A07).

Empirical research agenda point 4: Researchers should communicate clearly when and why an intervention showed room for improvement in terms of ST engagement, even if the effect was generally positive.

Fifth, the sample analyzed included clear examples of double and triple loop learning—e.g., embracing different practices in terms of sustainable communal living (A03), reflecting upon the past and envisioning new ways of moving forward through artistic practices (A06, A10), questioning pre-existing frames of reference and reflecting on community aspects through art (A14), reconfiguring epistemological positions away from neoliberal assumptions through affective and playful labor (A41), reflecting on and deeply changing practices in the professional setting (A52), and crafting new approaches to technological development and use (A83).

However, the full potential of these approaches, especially at the societal level, has yet to be entirely realized or understood across contexts. It is relevant to conduct more research exploring deeper forms of learning since ST gamification capable of affecting practices may require pursuing the transformational impact of double and triple loop learning (Van Mierlo and Beers, 2020), or at least questioning empirically what kind of learning is necessary for meaningful engagement with an ST (Stam et al., 2023; Subotticki et al., 2023). Beyond what the examples show, it remains to be seen to what extent gamification can trigger and support STs by changing frames of reference and situating learning in action, as is desirable in STs (Quang and De Wit, 2020; Van Mierlo and Beers, 2020).

As challenges, it is important to point out, first, that most of the interventions conducive to societal-level and triple loop learning did not use the words game or gamification. Hence, this represents a somewhat hidden cluster that traditional gamification designers and researchers can learn from, but they will need to understand and interpret conceptual and pragmatic differences—i.e., in ways to think but also in ways to communicate. Second, it must be acknowledged that triple loop learning is not only difficult to reach through an isolated gamified intervention (A24, A28), but it is also difficult to identify and measure (A24, A28). A clear understanding of how learning manifests and transparent methods are paramount for reliable conclusions.

Here, we propose our **empirical research agenda point 5: Gamified ST should target societal double and triple loop learning in action more often.**

Sixth, a minority of the outputs reported results related to the participants' experience, leading to the observation that emotions felt during play may be underexplored, in the same way that affective outcomes resulting from play are. Here, we advocate for a deeper and broader reporting of experiential outcomes, given their potential mediation of ST engagement. Thanks to experiential outcomes reported in this sample, we have not only identified intervention aspects leading to negative experiences, but also positive ones—e.g., the critical role of facilitation (A12, A74) and metaphors (A14) in ST learning; how playful environments can foster trust and connection (A37); and how making tasks emotionally involving can support meaning and motivation (A11).

As indicated throughout this review, a nuanced understanding of experience in ST research is both critical and challenging, given how ST engagement can be affected by aspects that become salient during play such as emotions (Martiskainen and Sovacool, 2021; Thelin, 2024; Valor et al., 2022), identity (Horsbøl, 2018), power dynamics (Avelino and Wittmayer, 2016), and autonomy (Ehnert et al., 2018). Since play experiences depend on the fit between the player, the context, and the activity, we recommend that researchers focus on studying the relationships between well-defined variables and outcomes on well-understood populations and contexts; that is, beyond an assumed general appeal of gamification. To address this, researchers may benefit from involving ethnographic methods (Metiu and Mukerjee, 2021; Rapp, 2017) or other qualitative approaches for a deeper understanding of user experience within STs, and how experience is afforded by gameful elements from the player's point of view. A key requirement is the need for researcher resources and expertise in conceptualizing and interpreting playful experiences (Masek and Stenros, 2021). Additionally, there is a need to create rapport with participants to establish honest communication, particularly when experiences are communicated rather than observed.

From this follows our **empirical research agenda point 6: Researchers should report and contextualize player experiences**

and their role in relation to ST engagement outcomes.

The six points above seek to advance empirical approaches to gamified ST by establishing connections with recommendations from ST research, play studies, and wider methodological and scientific concerns. This agenda aims to increase the quality and contribution of future studies by calling for more clearly defined participant samples in larger and longer studies (point 1); advocating for the use of comparisons even in the field (2); highlighting the need for further methodological variety (3) and a critical assessment of interventions (4); and calling for the exploration of deeper forms of learning (5) and a more complete reporting of player experiences in context (6).

4.4. Limitations

This scoping review includes both playful and game-like interventions for public engagement with STs. It therefore maps a broader space than reviews focusing only on games or gamification. The fact that we saw sufficient relevance in the playful interventions we analyzed suggests that in the future, search terms related to, e.g., interactive and participatory interventions may also prove fruitful in uncovering playful methods. However, for this review, which employs a systematic method and represents a first approach to this area, the search terms yield a sample that is commensurate with the purpose and our available resources.

Similarly, our search strategy focused on identifying interventions framed as part of STs, and therefore exhibiting awareness of a contribution beyond promoting sustainability in specific areas or contexts. Even so, our findings suggest that many gamified interventions focus on narrowly conceptualized sustainability changes, such as individual behavior, despite being explicitly framed as contributing to STs. This suggests that no clear-cut distinction can be made in terms of substance matter based on whether the intervention is framed as part of a transition or not. Here, since our interest has been to expand our existing knowledge in the direction of STs, we maintain that a search strategy leading to more studies adopting a generic sustainability framing (i.e., non-ST related) in areas such as energy, transport, or food would not have yielded further significant ST-specific insight, or at least commensurate with the effort required to analyze such a probably vast sample, as in fact these studies have been analyzed in previous reviews. In this sense, our findings and recommendations are likely applicable to the broader scope of gamified sustainability interventions.

Most importantly, though, our finding that framing may not be enough to demarcate the area of gamified ST interventions suggests an implication: should ST be a term applied liberally to any process promoting a more sustainable mindset or behavior, or should gamified ST interventions adopt specific ontological, epistemological, and methodological perspectives? Two avenues appear after this scoping review. Following an integrative path, future meta-reviews may bring together pre-existing foci on gamified intervention areas, topics, and contexts with ties to STs and analyze them through an explicitly ST lens. Following a more focused path, systematic literature reviews can be conducted with strict inclusion and exclusion criteria based on ST-related ontological, epistemological, and methodological aspects.

Finally, we acknowledge that some variables were especially sensitive to interpretation or analyst bias due to their complexity or lack of direct information. To mitigate this possibility, we screened every search result, and a minimum of two researchers extracted all data independently. Even so, the current study is a first step in an interdisciplinary area of research that will benefit from constant development and monitoring.

5. Conclusion

In a moment when abandoning incumbent systems in favor of more sustainable ones is crucial, play, games, and gamification have emerged as tools promising to promote stakeholder participation and cognitive, affective, and behavioral engagement with sustainability issues (Fernández Galeote et al., 2021). In connection with central ST research concerns, we must ask whether gamification can transcend localized and short-term sustainability interventions towards (a) transforming practices and (b) supporting the integration of psychological and behavior changes into larger transition dynamics.

This scoping review has followed a systematic process to uncover how play, games, and gamification have been used to support STs. While the two questions above should continue to be explored in more focused studies, reviews, and meta-reviews compiling pre-existing foci on gamified intervention areas, topics, and contexts with ties to ST, our findings provide novel insights relevant to them. First, regarding whether and how gamification is used to transform practices, it appears that a focus on supporting user agency is typical, and many interventions prefer to emphasize education, simulations, and agency-focused behavior change rather than address practices in their full complexity. Second, regarding whether and how agentic changes (e.g., in awareness, motivation, or behavior) influence larger transition dynamics, we found only a few large-scale interventions in terms of sample size, time, and space, but sufficient variety to inspire multiple future avenues. Although a minority, multiple interventions reported double and triple loop learning outcomes, and over half reported some form of direct real-world impact. More generally, the sample included multiple examples in which play, game, and gamified activities were part of larger interventions.

Thus, while notable attempts are being trialed and implemented, more work is needed to develop an ecology of STs that incorporates gamification and explores and leverages its capacity to alter ST environments and motivate and gather diverse actors, and in this way support practice and large-scale transitions. To advance this goal, we have presented three different agendas. First, our contextual agenda aims to enhance the studies' alignment with their targeted transitions by proposing more explicit and diverse conceptual frameworks, including a broader range of practitioners, considering the role of affect, and embedding gamified interventions within direct socio-technical and socio-ecological transitions. We also propose implementing these interventions beyond the West and at larger scales. Second, the design and implementation agenda focuses on the conceptualization, development, and application of interventions. We advocate for future studies to increase the effectiveness and diversity of designs through effective user-

centered design strategies, the consideration of individual player needs, stronger design justifications, playful designs beyond traditional games and singular goal structures, the use of immersive elements and technologies, and the addition of pre-existing sustainability and engagement techniques. Third, our empirical research agenda aims to increase the quality and contribution of future studies by calling for more clearly defined participant samples in larger and longer studies, including comparisons even in the field, further methodological variety, criticality in intervention assessment, and calling for the exploration of deeper forms of learning and a more complete reporting of player experiences in context. If these recommendations are considered, future research should increase its rigor and impact, thereby advancing toward a more meaningful understanding and application of gamified approaches in support of sustainability transitions.

CRediT authorship contribution statement

Daniel Fernández Galeote: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Linas Gabrielaitis:** Writing – review & editing, Methodology, Investigation. **Georgina Guillén:** Writing – review & editing, Methodology, Investigation. **Juho Hamari:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.eist.2025.101025.

Data availability

The research data has been included as a supplemental file.

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